



By Ivan Revin

**for MIT GLOBAL AI
HACKATHON 2025**

Introduction

Hi, everyone!

My name is Ivan Revin.

I am 10th grade student of Iprep Academy North from Miami-Dade, Florida. I am interested in programming in Python and P5JS, artificial intelligence and its application in everyday life, as well as the development of computer games.



Purpose of the Presentation

The goal of this presentation is to show you a cool app is developed to help students with their science studies. With all the different school standards in the U.S., it can be tough to find the right resources. This app is designed to bring everything together, making studying easier with practice tests. I want to show how this tool can make learning more effective and boost students' confidence in their schoolwork.





Background

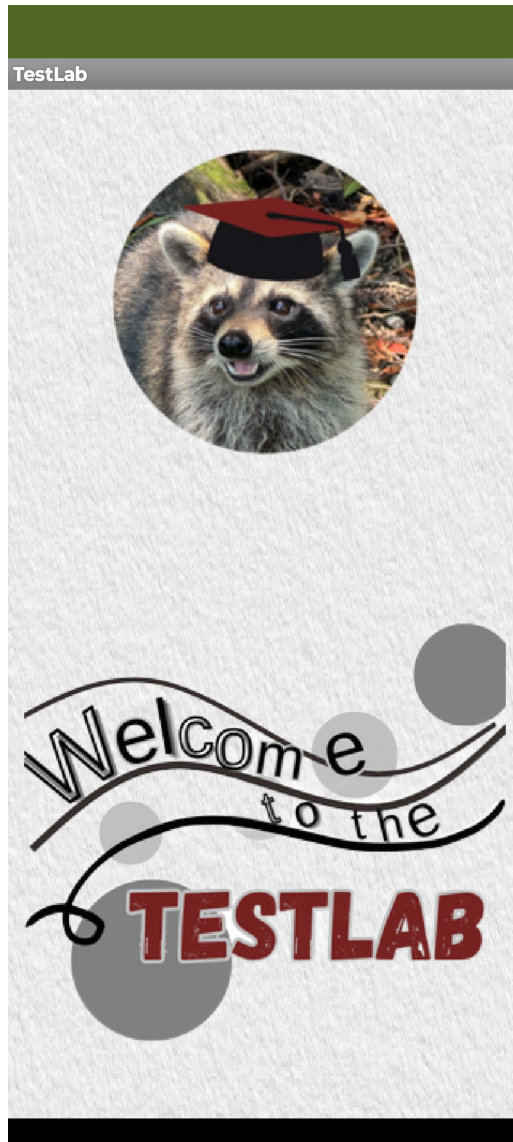
Problem Statement

In U.S. schools, one big issue is that every state has its own education standards. This makes it tough to have the same curriculum and assignments everywhere. Students are left trying to keep up with different textbooks and materials without one reliable place to study. Plus, a lot of the resources out there just give info without interactive practice that really matches what we learn in class.

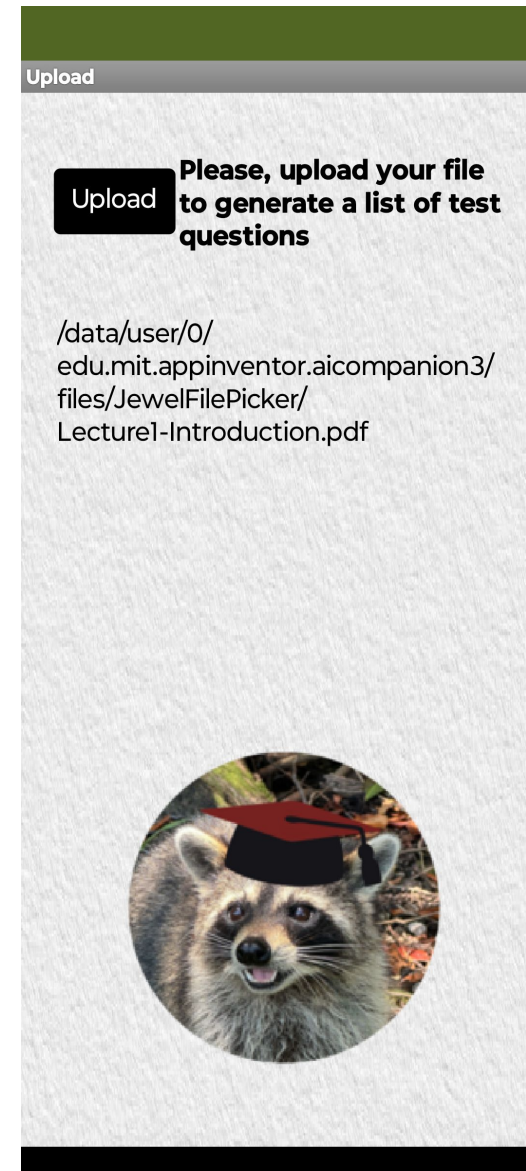
Objective of the App

My app is here to fix that! It's designed to help students with natural sciences by bringing everything together in one place. You can use it to take practice tests that suit your needs. It's all about making learning easier and more personal, so students can understand better and do well in school, college and university.

App Overview

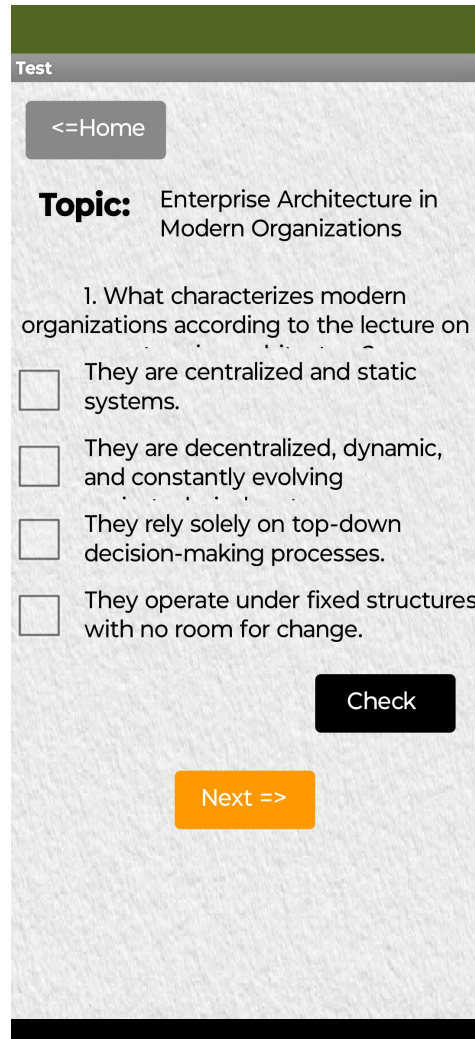


The first screen appears as a **welcome** of the app. It serves to introduce users to "TestLab" and creates a friendly, engaging start with the playful raccoon image.

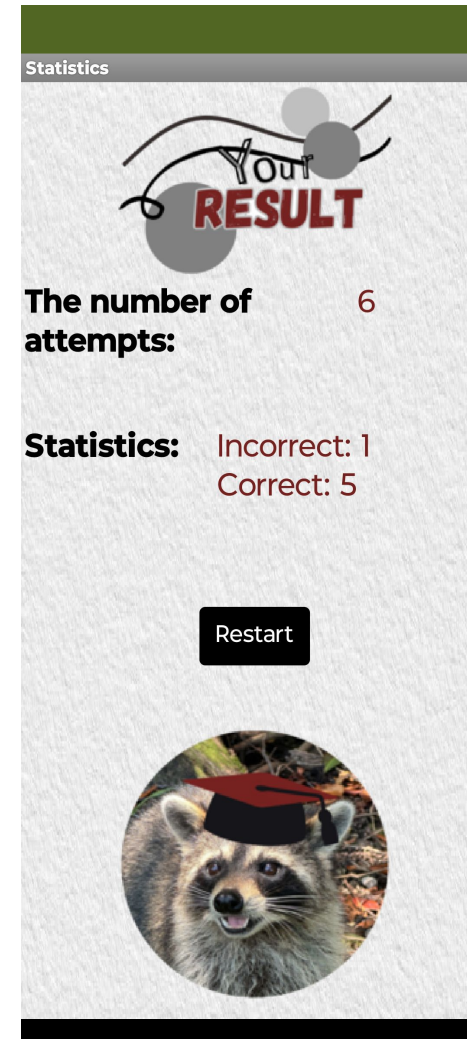


This screen guides users through uploading their study materials to generate customized test questions. At the top, it highlights the 'Upload' section. The convenient upload button makes it easy to select and submit files. Once a file is chosen, its path is displayed, confirming successful selection.

App Overview - 2



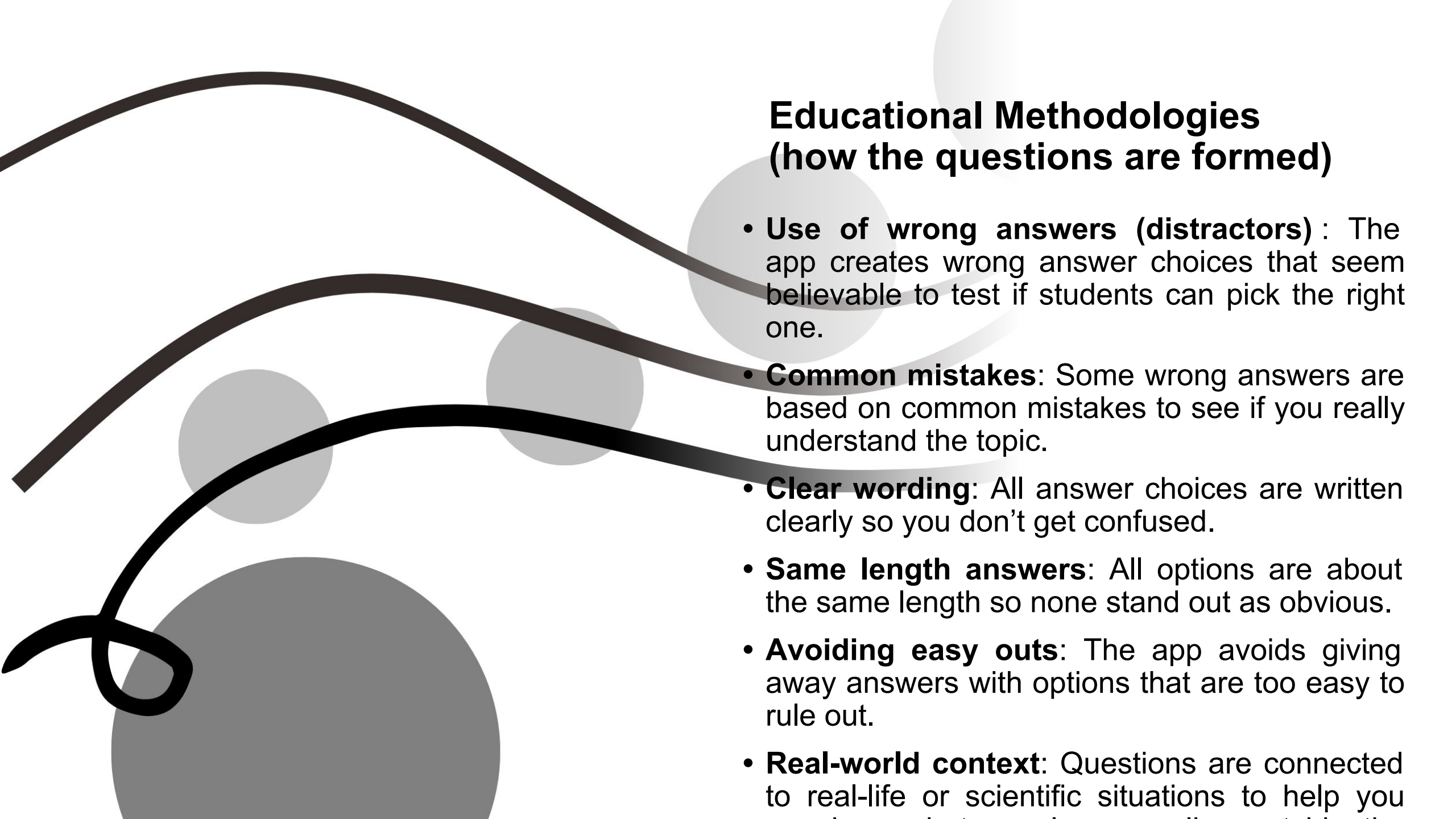
This screen is where users engage with the testing feature of the app. Users can easily return to the main menu using the '<=Home' button. The test begins by displaying the topic. Users are given four multiple-choice options, each with a checkbox for easy selection. Once an answer is chosen, they can click 'Check' to submit and receive feedback, and the 'Next' button allows them to move seamlessly to the following question if it was chosen the right answer.



This screen provides users with a overview of their performance. Below, 'Your Result' title displays the total number of attempts. The statistics show a detailed breakdown of outcomes, with both correct and incorrect answers listed. A 'Restart' button encourages users to improve by retaking the test.

Target Subjects

The subjects the app can assist with, such as biology, history, and other non-mathematical subjects focused on definitions and terms.



Educational Methodologies (how the questions are formed)

- **Use of wrong answers (distractors)** : The app creates wrong answer choices that seem believable to test if students can pick the right one.
- **Common mistakes**: Some wrong answers are based on common mistakes to see if you really understand the topic.
- **Clear wording**: All answer choices are written clearly so you don't get confused.
- **Same length answers**: All options are about the same length so none stand out as obvious.
- **Avoiding easy outs**: The app avoids giving away answers with options that are too easy to rule out.
- **Real-world context**: Questions are connected to real-life or scientific situations to help you

Benefits

The app comes with great perks for students. You can upload any information from your teacher—whether it's from textbooks, presentations, or articles (just convert it into .pdf)—and the app helps transform it into practice tests. This way, you can focus on what you need to study, making learning more targeted and helping you improve your grades. It's all about making your study time more effective!



**Your
RESULT**



Conclusion

To wrap it up, we've talked about the challenges in U.S. school education with different standards and how this app helps students by providing customized practice tests using their class materials. It simplifies studying and boosts understanding of natural sciences.